Giacomo Veneri

MEng, PhD

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Summary

24 years **Data Architect and AI/IoT Specialist** focusing on **IoT**, **Cloud and Artificial Intelligence**. **7 years as AI Specialist** working on hybrid data model, generative model, deep learning model applied to Finance, Automotive, Industry and Healthcare. 13 years people leader. 2 Books, 30+ articles. 7 Certifications, 10 Awards

Work experience

Director Artificial Intelligence Specialists — January 2022 - now

Baker Hughes (Energy, 55K employees)

Developing Artificial Intelligence solutions for Advanced Manufacturing and IIoT

- Define the internal roadmap for AI
- Peoples Leader (20 internal + 3 external + 2 PhD)
- Budget Management & Roadmap definition
- Developing Deep Learning solutions in the field of IoT, Computer Vision and NLP/LLM
- Developing ML processing solutions and pipeline

Achievements: 20+ ML projects delivered, 50+ POC delivered, team leader of 20+ peoples, 5+ patents delivered

Methodology: Agile and Scaled Agile Management, Kanban, Lean Agile

Senior Artificial Intelligence Specialist — December 2018 - January 2022

Baker Hughes (Energy, 56K employees)

Developing Artificial Intelligence solutions for Advanced Manufacturing and IIoT

- Developing Deep Learning solutions in the field of NLP/LLM and Computer Vision
- Developing ML processing solutions and pipeline
- Developing IoT analytics: Predictive Maintenance, Anomaly Detection, Production forecasting and optimisation, Inventory Optimisation
- Building Cloud Infrastructure
- Define the internal roadmap for AI.
- Small team (5 peoples) leader

Achievements: 10+ ML projects delivered, 3 Production ready solutins. C3.Ai Fundamentals certified

Technologies: Docker, Kubernetes, Microservices, AWS, DynamoDB, EC2, IOT, AWS Lambda, Python, Cassandra, IOT, Machine Learning , Git, Rally, Confluence, Keras, TensorFlow 2, PyTorch, Tranformers, OpenCV, GPU NVIDIA, Scikit-Learn, Dask, Parquet, HDFS, OpenTSDB, Django, MLFlow, Imblearn, Hugging Face, Scikit-Image, GluonTS, Retinanet, VGG16, Inception V3, ResNet, Mask R-CNN, Shapely, LIME, ML Classic (XGB, RadomForest, K-NN, HDBScan, PCA, U-MAP, DTW, TF-IDF ..,)

AI & IoT Technical Manager — May 2017 - December 2018

Baker Hughes a GE Company (Energy, 320K employees)

Developing IOT and AI digital product in the field of Oil & Gas

- Product's feature definition and validation
- IT Architecture definition
- Scrum Product Management
- Big Data Analytics definition and proposal
- Machine, Digital Twins and Physics Based analytics
- Support the development of Value Proposition
- UX Design validation and user's story proposal
- Project Management, Agile, Lean, SCRUM (2) Team coordination

Achievements: Working on 2 projects and 2 products in the field of Oil & Gas as Technical Product Manager & Sw Architect. NVIDIA Institute certified

Skills

Artificial Intelligence

Deep Neural Network, LLM, DPO, CNN, Transformer, RNN, Computer Vision, xVAE, Regression Analysis, Explainable AI, **NLP** and Topic Extraction, ML with Constraints, Generative AI, Adversarial NN for Domain Adaptation, Stacking Methods, Unbalanced Learning, Timeseries Forecasting, Classification, Attention Model, Python, Keras, Tensorflow, Scikit-Learn, OpenCV, **GPU Nvidia** DGX, AWS SageMaker, **C3.AI,** Transformers, Normalising Flow

Software Architect

Spring, OSGI, Karaf, Angular, **Java**, JEE, JPA, JNA, Oracle Weblogic, WebSphere, JBOSS, Tomcat, Oracle DB, Oracle Spatial, Oracle PL/SQL, RichFaces4, Angular, AngularJS, EJB, SOAP, JMS, UML 2.0, .NET, WPF, WCF, ASP.NET, SQL Server, Visual C++, Android, Maven, MVC, Apache, Ngnix, ETL, Talend, Play, Bower, NodeJS, Grunt, Gulp, Bootstrap, SASS, PostgreSQL, MySQL, Pattern Programming, Flask, FastAPI

Cloud Infrastructure

Cloud Foundry, AWS, AWS EC2, AWS IoT, Azure IoT, GCP IoT, AWS ELB, **Predix** Cloud,Docker, Kubernetes, AWS SageMaker, AWS Textract, AWS DynamoDB, AWS Lambda, Azure Functions, Azure Insights, Azure PowerBI, Azure ML, AWS Greengrass, AWS Elasticsearch, AWS S3, AWS EMR

Agile Develoment

SCRUM, Lean Development, XP, Kanban, Continuous Integration, TDD, **SAFE**

Big Data

Cassandra, Spark, HBase, Hadoop, Sqoop, Microservices, RabbitMQ, Redis, Docker, GE Historian, ElasticSearch, OpenTSDB, KairosDB, Solr, HiveQL, Parquet, ORC, Kafka

Product Management

Technical Product Management, UX Definition, Features Definition, Software Lifecycle Management, **Green Belt**

Technologies: Docker, Kubernetes, AWS, DynamoDB, EC2, IOT, Google Big Table, Azure Data Lake, data Lake Analytics, AWS Lambda Predix, Microservices, Java 8/9, CF, Spring, Spring Cloud, Spring Security, Python, Cassandra, Elasticsearch, Kibana, M2M, IOT, SCADA, Kafka, RabbitMQ, Machine Learning , Polymer, Angular, Maven, Git, Github, JIRA, Rally, AHA!, Jenkins, CheckMarx, Rest, SOAP, Solr, Hive, HYSYS, PetroSIM, Scikit-Learn, Tensorflow

AI & IoT Technical Leader — Dec 2014 - May 2017

General Electric Oil & Gas (Global,320K employees)

Developing Advanced Analytics project and IoT infrastructure

- IT Architecture definition
- Building data acquisition and data processing projects in the filed of AI and I-IoT; developing architecture able to consume big amount of data (time series).
- Analytics and AI development
- Analytics Data Platform development
- Team coordination (SCRUM Master)
- Small team (2-4 peoples) people leader

Achievements: 5 International projects in the field of Oil & Gas as Team and Technical Leader and Senior Architect. Predix Builder Influencer. Predix Cloud certified. GB certified, SCRUM certified

Technologies: OSGI, AWS, EC2, IOT, Predix, Microservice, Oracle, PostgreSQL, Java 7/8, CF, Spring, Python, Cassandra, Elasticsearch, Kibana, Logstash, Historian TS, TSDB, Hadoop, Redis, M2M, IOT, SCADA, OpenTSDB, KairosDB, Solr, HiveQL, Parquet, ORC, Kafka, RabbitMQ, Machine Learning , Polymer, AngularJS, Splunk, Maven, Git, Github, JIRA, Rally, AHA!, Jenkins, Nagios, CheckMarx, Rest, SOAP, ETL, Nginx, Flink, , Scikit-Learn, Python

Team Leader — Sep 2012 - Dec 2014

Engineering Ingegneria Informatica SpA (IT, 7K employees)

Development of IoT Automotive System

- Developing of IoT and Machine Tracking projects
- Near Realtime System development
- Test driven development
- CI/CD Support
- Tolling component functional analysis, SOA Integration (SAP-FI)
- Small team (3-5 peoples) people leader

Achievements: 6 International projects in the field of Automotive as Team Leader and Senior Architect. Introduction of CID and TDD. Java OCA Certified. 1 Book.

Technologies: Weblogic, Analytics, IoT, JMS, EJB, Websphere, SpagoBI, Oraccle Spatial, GIS, JBOSS, Oracle, PostgreSQL, MySQL, Java 6/7, Spring, PHP, Hadoop, C/C++, Android,JSF, AngularJS, Javascript, Maven, Git, Github, JIRA, Jenkins, Sonar, FindBugs, Rest, SOAP

Chief Technology Officer — Jan 2007 - Sep 2012

Etruria Innovazione SCpA (IT, 16 employees)

Director of MIUR Human Computer Interaction Laboratory

- Developing digital projects about Business Intelligence for bank
- Developing eProcurement and Machine Learning projects for PA
- Design and development of digital solutions
- Preliminary design, feasibility and execution
- Definition of the software development life cycle (SDLC) using Agile methodology.
- Research development in the field of NLP, healthcare and neuroscience
- Team (8 peoples) people leader

Achievements: Coordinating 2 european projects. More than 30 Regional/national projects in the field of R&D. Promoting the ISO: 9001 and the accreditation of the laboratory to the MIUR of development that I direct.

Technologies: Weblogic, JMS, EJB, , Webshpere, JBOSS, Python, Weka, Tomcat, Oracle, PostgreSQL, MySQL, Java 5/6/7, Spring, PHP, C/C++, JSF, Javascript, Maven, Git, SVN, C#, .NET, SQL Server, ASP.NET, ASP, SOAP, WPF, WCF

Senior Developer — Jan 2006 - Dec 2006

ION Trading srl (IT, 1K employees)

Research and Development of Financial Products

- Java and C/C++ developer

Statistics

Bayesian Model, Monte Carlo Model, SPSS, Matlab, Inferential Statistics

Languages

- English : B2
- French : A2
- Italian : mother tongue

Awards

1. NASSCOM 2023
2. 7th Best IoT World Book - Nov 2019 by BookAuthority
3. 1st AWS Deepracer Competition - Nov 2019 by AWS
4. We are inventors - Dec 2018 by BHGE
5. Customers determine our success - Sep 2016 by GE Oil & Gas
6. Predix Builder Influencer - Jul 2016 by GE Digital
7. Empower and inspire each other - Jan 2016 by GE Corporate
8. Learn and adapt to win - Nov 2015 by GE Oil &Gas
9. European Network - Feb 2010 by European Community
10. Dire & Fare - Feb 2008 by Regione Toscana

Patents

Device for the analysis of the central nervous system by application of a combination of stimuli and the study of corresponding reactions Rufa, Veneri, Federico, Federighi, Santarnecchi IT patent pct/it2010/000,300

Research Activities

1. Collaboration with University of Siena School of Medicine in the field of Neurology and Biomedical Systems and School of Engineering
2. Collaboration with University of Siena Engineering Artificial Intelligence Lab
3. Collaboration with Pennsylvania University - PINNs
4. Artificial Intelligence and IIoT with University of Sydney
5. Referee for IEEE / ACM Transactions on Computational Biology and Bioinformatics

Social

- Research Gate
- StackOverflow
- LinkedIn
- Amazon Author
- Scholar

- Research and development of the middleware platform of ION Trading (MktView).
- developing solutions for managing security and access rights to the data
- introducing new encryption technologies, policy management and licensing management
- increasing performance and reduce latency data transmission and communication between processes

Technologies: Java, C/C++, Bloomberg, MarketView, .NET, C#

Chief Technology Officer — 2004 - 2006

Etruria Innovazione SCpA (IT, 16 employees)

Research and Development of Knowledge Systems

- Research & development of Digital Projects, Machine Learning and NLP Solutions.
- Preliminary design, feasibility and execution.
- Definition of the software development life cycle (SDLC) using Agile methodology
- Team (6 peoples) people leader

Achievements: 7 Regional/national projects in the field of PA. 12 products for PAs and Banks.

Technologies: Weblogic, JMS, EJB, , JBOSS, Tomcat, Oracle, PostgreSQL, MySQL, Java 5/6, Spring, PHP, C/C++, JSF, Javascript, ANT, SVN, C#, .NET, SQL Server, ASP.NET, ASP, TF-IDF, NLP, Semantic NLP, CNL, Regression, Tree Classifier

President, Co-Founder & Team Leader — Jan 2001 - Dec 2004

Metaresearch srl (IT, 6 employees)

Research and Development of NLP and Knowledge management products

- Developing and designing the solution and / or products in the field of NLP and Knowledge Management
- Developing natural language analytics.
- Collaboration with the University of Siena for the development of intelligent systems based on genetic algorithms and text mining
- Collaboration with CNR of Pisa Department of Computational Linguistics to develop NLP solutions
- Small team (3 peoples) people leader

Technologies: Websphere, JMS, EJB, JBOSS, Tomcat, Oracle, PostgreSQL, MySQL, Java 5/6, Spring, PHP, C/C++, JSF, Javascript, ANT, SVN, C#, .NET, SQL Server, ASP.NET, ASP, Visual C++, WordNet, LISP, TF-IDF, NLP, Semantic NLP, CNL

Consultant Engineer — 1999 - 2005

University of Siena

Research and Development of Multimedia and Computer Vision Systems

- Development of navigation software for multimedia, high-resolution images of large (> 3GB).
- Development of Electrophoretic Software analysis based on Computer Vision

Technologies: Tomcat, Oracle, PostgreSQL, MySQL, Java , C/C++, JSF, ANT, SVN, SQL Server, ASP, Visual C++, TF-IDF, NLP, Semantic NLP

Doctorate - PhD — 2008 - 2012

Università degli Studi di Siena

Bioinformatic and Neuroscience

Thesis : Feature-based information processing of selective attention

Engineer Master's Degree - MSc — 1992 - 1999

Università degli Studi di Siena

Telecommunication Engineer

Thesis: Image Processing

Certifications

SAFe Pratictitioner 6 — 2023

SAFe

Agile SAFE Certified

Last 10 Publications

Controllable Image Synthesis of Industrial Data Using Stable Diffusion - Valvano, Agostino, De Magistris, Graziano, G. Veneri; WACV, 2024

Combining Thermodynamics-based Model of the Centrifugal Compressors and Active Machine Learning for Enhanced Industrial Design Optimization, Ghiasi, Pazzi, Del Grosso, De Magistris, Veneri, SynS & ML ICML2023

Sensor virtualization for anomaly detection of turbo-machinery sensors - An industrial application
Shetty, Gori, **Veneri**
9th International conference on Time Series and Forecasting - ITISE2023

Document Layout Analysis with Variational Autoencoders: An Industrial Application, Youssef, Valvano, **Veneri**, **ISMIS** 2022. Lecture Notes in Computer Science, 2022

Continual Learning for anomaly detection on turbomachinery prototypes - A real application, Gori, **Veneri**, Ballarini, 2022 **IEEE** CEC 2022

Deep Surrogate of Modular Multi Pump using Active Learning
Murugesan, Goyal, Barriere, Pasquotti, **Veneri**, De Magistris
Adaptive Experimental Design and Active Learning in the Real World - **ICML** 2022

DANNTe: a case study of a turbo-machinery sensor virtualization under domain shift. Strazzera, Gori, **Veneri**, **NeurIPS** 2021 Workshop DistShift, Dec 2021

Learning to Identify Drilling Defects in Turbine Blades with Single Stage Detectors Panizza, Stefanek, Melacci, **Veneri**, Marco Gori, Machine Learning for Engineering Modeling, Simulation, and Design Workshop at **NIPS** 2020

Veneri and Capasso, Hands On Industrial Internet of Things, Packt, 2018, **Book**

Maven Build Customization, Packt, 2014, Anardu, Baldi, Cicero, Giomi, Veneri, **Book**

Militar Service

Required Military Service full-filled.
Mar 2001- Dec 2001

4 Projects

(Public Resources)

Natural Language processing Specialization 2021
DeepLearning.ai
NLP Specialization

C3 Fundamentals 2020
C3.ai
Data Lake Management.

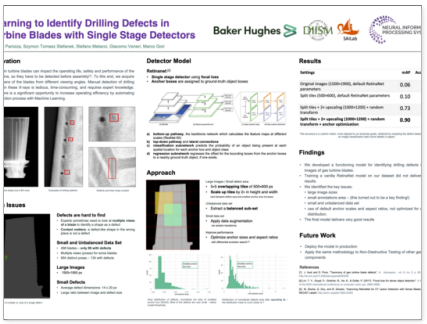
DEEP LEARNING FOR INTELLIGENT VIDEO ANALYTICS 2019
NVIDIA
AI Development

Oracle certified 1Z0-803 – OC1280222 2012
Oracle University
Java 7 Certified

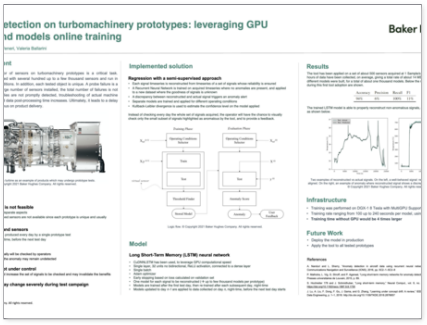
Green Belt Certified 2017
Six Sigma
Lean Project Management and Quality Certification

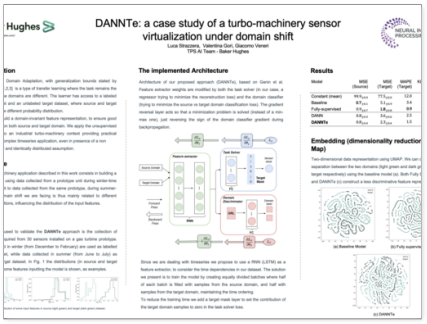
Scrum Fundamentals Certified 2017
SCRUM STUDY
SCRUM and Agile

Predix Developer Certified 2017
General Electric
Predix Cloud certification, Cloud Foundry, Microservices, Spring Boot

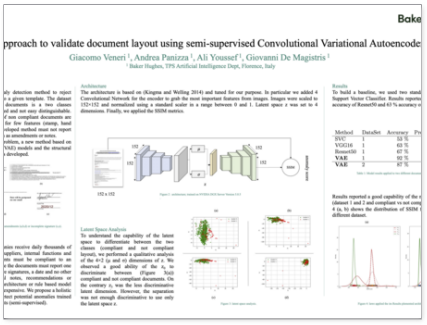


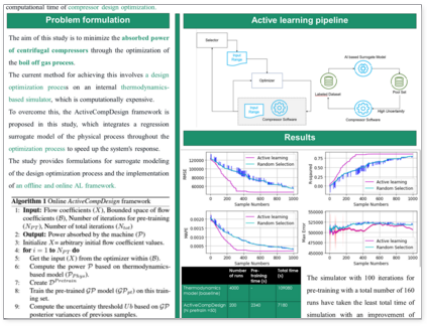
Defect Detection


Anomaly Detection



Domain Adaptation


Document Layout Analysis



Active Learning

Created with VisualCV